

St George Girls High School

Year 12

Common Test 3

2009



Mathematics Extension 1

General Instructions

- Working time – 75 minutes
- Reading time – 5 minutes
- Write using blue or black pen
- Board-approved calculators may be used.
- A table of standard integrals is provided.
- All necessary working should be shown in every question.
- Write on one side of the page only.
- Start each question on a new page.

Total marks – 66

- Attempt Questions 1 – 6
- All questions are of equal value

Question	Mark
Question 1	/11
Question 2	/11
Question 3	/11
Question 4	/11
Question 5	/11
Question 6	/11
Total	/66

Students are advised that this is a School Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

Question 1 – Start a New Page – (11 marks)

Marks

a) If α, β, γ are the zeros of the polynomial $P(x) = 2x^3 + 8x^2 - x + 6$ evaluate:

(i) $\alpha + \beta + \gamma$

1

(ii) $\alpha\beta + \beta\gamma + \alpha\gamma$

1

(iii) $\alpha\beta\gamma$

1

(iv) $\alpha^2 + \beta^2 + \gamma^2$

2

b) By drawing a suitable graph, solve $x^2(x - 1)(x + 2) \geq 0$

2

c) When $Q(x) = ax^3 + bx^2 + c$ is divided by $(x + 2)$ the remainder is 3 and, when $Q(x)$ is divided by $(x^2 - 1)$ the remainder is $(2x + 4)$.

4

Find a, b and c .

Question 2 – Start a New Page – (11 marks)

Marks

a) Evaluate: $\int_0^{\frac{\pi}{3}} \sin^2 2x \, dx$ 2

b) The polynomials $3x^3 - x + 1$ and $ax(x - 1)(x + 2) + bx(x - 1) + cx + d$ are equal for ~~all~~ ^{all} values of x . 4

Determine the values of a , b , c and d .

c) (i) Express $2 \sin x + \sqrt{12} \cos x$ in the form $R \sin(x + \theta)$ where $R > 0$ and θ is a subsidiary angle in the range $0 \leq \theta \leq \frac{\pi}{2}$ 5

(ii) Hence, give the general solution to the equation

$$2 \sin x + \sqrt{12} \cos x = 2\sqrt{3}$$

Question 3 – Start a New Page – (11 marks)

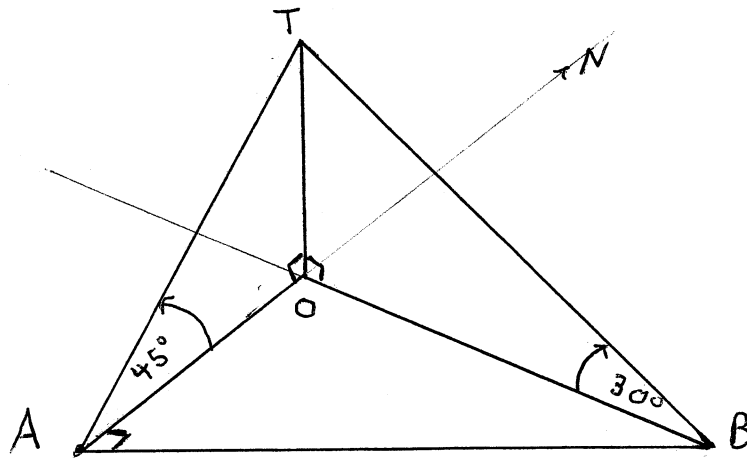
Marks

a) Evaluate: $\int_0^{\frac{1}{4}} \frac{dx}{\sqrt{1-4x^2}}$ 2

b) Sketch the graph of $y = \sin^{-1}\left(\frac{x}{2}\right)$. Clearly indicate the domain and range on your graph. 3

c) Differentiate $(1+x^2) \cdot \tan^{-1} x$ 2

d) 4



A surveyor stands at point A due south of a tower OT of height h metres.

The angle of elevation of the top of the tower from A is 45° . The surveyor then walks 100 metres due east to point B , from here the angle of elevation to the top of the tower is 30° .

(i) Show that $h = 50\sqrt{2}$

(ii) Calculate the bearing of B from the base of the tower.

Question 4 – Start a New Page – (11 marks)

Marks

- a) Evaluate: $\sin^{-1}\left(\cos\frac{\pi}{3}\right)$ 2
- b) Use the table of standard integrals to show that $\int_0^{15} \frac{dx}{\sqrt{x^2+64}} = 2 \ln 2$ 2
- c) (i) Show that $\frac{u}{u+1} = 1 - \frac{1}{u+1}$ 1
- (ii) Hence, find $\int \frac{dx}{1+\sqrt{x}}$ using the substitution $x = u^2$ [$u \geq 0$] 3
- d) Given that $y = \sin^{-1}(\sqrt{x})$, show that $\frac{dy}{dx} = \frac{1}{\sin 2y}$ 3

Question 5 – Start a New Page – (11 marks)

Marks

- a) Show that $(2x + 1)$ is a factor of $2x^3 + 7x^2 - x - 2$ 1
- b) (i) Show that $\sin x - \cos 2x = 2\sin^2 x + \sin x - 1$ 2
- (ii) Hence, or otherwise, solve $\sin x - \cos 2x = 0$ for $0 \leq x \leq 2\pi$ 3
- c) If $a \cdot \cos x = 1 + \sin x$
- (i) prove that $\frac{a-1}{a+1} = t$, where $t = \tan \frac{x}{2}$ 3
- (ii) hence, solve $1 + \sin x = 2 \cos x$ for $0^\circ < x < 360^\circ$, to nearest degree. 2

Question 6 – Start a New Page – (11 marks)

Marks

- a) Evaluate: $\int_1^{e^3} \frac{dx}{x(9+(\ln x)^2)}$ using the substitution $u = \ln x$ 4
- b) A monic polynomial $P(x)$ of degree 4 is known to have exactly two zeros at 2 and -2 . It is also known that $P(x)$ is an even function. 3
- Further, when $x = 3$ the value of $P(x)$ is 55. Determine the polynomial function $P(x)$.
- c) Consider $\tan^{-1}y = 2\tan^{-1}x$. 4
- (i) Express y as a function of x
- (ii) Show that the function has no turning point.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

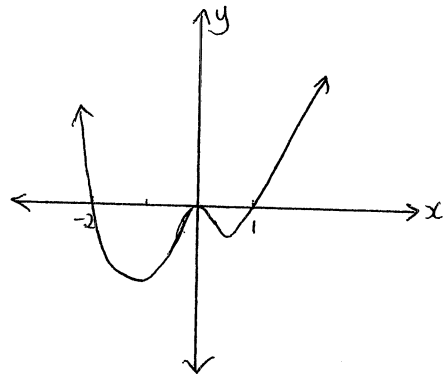
Question 1

a) $P(x) = x^3 + 8x^2 - x + 6$

$$\begin{aligned} \text{(i)} \quad x + y + z &= -\frac{b}{a} \\ &= -\frac{8}{1} \\ &= -8 \\ \text{(ii)} \quad x + y + z &= \frac{c}{a} \\ &= \frac{6}{1} \\ &= 6 \\ \text{(iii)} \quad x + y + z &= -\frac{d}{a} \\ &= -\frac{-1}{1} \\ &= 1 \end{aligned}$$

$$\begin{aligned} \text{(iv)} \quad x^2 + y^2 + z^2 &= (x + y + z)^2 - 2(xy + xz + yz) \\ &= (-8)^2 - 2(-1) \\ &= 64 + 2 \\ &= 66 \end{aligned}$$

b)



$x \leq -2$, $x \geq 1$ and $x = 0$

c) $Q(x) = ax^3 + bx^2 + C$

$Q(-2) = 3$

$$\begin{aligned} a(-2)^3 + b(-2)^2 + C &= 3 \\ -8a + 4b + C &= 3 \quad \text{①} \end{aligned}$$

$$\begin{aligned} \text{②} - \text{③} \quad & a + b + C = 6 \quad \text{②} \\ \text{we get} \quad & -a + b + C = 2 \quad \text{③} \end{aligned}$$

$$\begin{aligned} 2a &= 4 \\ \therefore a &= 2 \end{aligned}$$

Subst. $a = 2$ into ① we get

$$\begin{aligned} -8(2) + 4b + C &= 3 \\ -16 + 4b + C &= 3 \\ 4b + C &= 19 \quad \text{④} \end{aligned}$$

④ - ⑤ we get

$$\begin{aligned} 4b + C &= 19 \\ b + C &= 4 \end{aligned}$$

$$\begin{aligned} 3b &= 15 \\ \therefore b &= 5 \end{aligned}$$

So $a = 2$, $b = 5$, $C = -1$

$$\begin{aligned} Q(1) &= 2 + 4 = 6 \\ Q(-1) &= -2 + 4 = 2 \\ a(1)^3 + b(1)^2 + C &= 2 + 4 \\ a + b + C &= 6 \quad \text{②} \\ a(-1)^3 + b(-1)^2 + C &= 2 \\ -a + b + C &= 2 \quad \text{③} \end{aligned}$$

Subst. $a = 2$ into ② we get

$$\begin{aligned} 2 + b + C &= 6 \\ \therefore b + C &= 4 \quad \text{⑤} \end{aligned}$$

Subst. $b = 5$ into ⑤ we obtain

$$\begin{aligned} 5 + C &= 4 \\ \therefore C &= -1 \end{aligned}$$

Question 2

$$a) \int_0^{\frac{\pi}{3}} \sin^2 2x \, dx$$

$$= \int_0^{\frac{\pi}{3}} \frac{1}{2} (1 - \cos 4x) \, dx$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{3}} (1 - \cos 4x) \, dx$$

$$= \frac{1}{2} \left[x - \frac{\sin 4x}{4} \right]_0^{\frac{\pi}{3}}$$

$$= \frac{1}{2} \left\{ \left[\frac{\pi}{3} - \frac{1}{4} \sin \frac{4\pi}{3} \right] - \left[0 - \frac{1}{4} \sin 0 \right] \right\}$$

$$= \frac{1}{2} \left\{ \left[\frac{\pi}{3} - \frac{1}{4} \times -\frac{\sqrt{3}}{2} \right] - [0] \right\}$$

$$= \underline{\underline{\frac{1}{2} \left[\frac{\pi}{3} + \frac{\sqrt{3}}{8} \right]}}$$

$$b) 3x^3 - x + 1 \equiv ax(x-1)(x+2) + bx(x-1) + cx + d$$

Equating

$$\text{coeff of } x^3 \quad 3 = a$$

$$\text{when } x=0 \quad 1 = d$$

$$x=1 \quad 3 = c + d$$

$$\text{so } 3 = c + 1$$

$$\therefore c = 2$$

Try when
 $x = -2$,

$$3(-2)^3 - (-2) + 1 \equiv 0 + b(-2)(-2-1) + c(-2) + d$$

$$-24 + 2 + 1 \equiv 6b - 2c + d$$

$$-21 \equiv 6b - 2 \times 2 + 1$$

$$-21 \equiv 6b - 4 + 1$$

$$-21 \equiv 6b - 3$$

$$6b \equiv -18$$

$$\therefore b = -3$$

so

$$\underline{\underline{a=3, \quad b=-3, \quad c=2, \quad d=1}}$$

(i) $2\sin x + \sqrt{12}\cos x$

$R \sin(x+\theta)$, $R > 0$

$$R \sin(x+\theta) = R \sin x \cos \theta + R \cos x \sin \theta$$

so $2\sin x + \sqrt{12}\cos x = R \sin x \cos \theta + R \cos x \sin \theta$

Equating
coefficients of
 $\sin x$,

$$R \cos \theta = 2 \quad (1)$$

Equating
coefficients of
 $\cos x$

$$R \sin \theta = \sqrt{12} \quad (2)$$

Squaring
and adding
(1) and (2)

$$R^2 \cos^2 \theta + R^2 \sin^2 \theta = 4 + 12$$

$$R^2 (\sin^2 \theta + \cos^2 \theta) = 16$$

$$\therefore R^2 = 16$$

$$R = \pm \sqrt{16}$$

but $R > 0$

$$\text{so } \underline{R=4}$$

From (1),

$$4 \cos \theta = 2$$

$$\cos \theta = \frac{1}{2}$$

$$\therefore \theta = \frac{\pi}{3}$$

From (2)

$$R \sin \theta = \sqrt{12}$$

$$4 \sin \theta = \sqrt{4}\sqrt{3}$$

$$4 \sin \theta = 2\sqrt{3}$$

$$\sin \theta = \frac{\sqrt{3}}{2}$$

$$\therefore \theta = \frac{\pi}{3}$$

so θ is in the first quadrant with related angle $\frac{\pi}{3}$

$$\text{so } 2\sin x + \sqrt{12}\cos x = 4 \sin \left(x + \frac{\pi}{3}\right)$$

(ii) $2\sin x + \sqrt{12}\cos x = 2\sqrt{3}$

then $4 \sin \left(x + \frac{\pi}{3}\right) = 2\sqrt{3}$

$$\sin \left(x + \frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

$$\sin \left(x + \frac{\pi}{3}\right) = \sin \frac{\pi}{3}$$

ie

$$x + \frac{\pi}{3} = 2n\pi + \frac{\pi}{3}$$

$$\text{or } x = (2n+1)\pi - \frac{\pi}{3}$$

$$\underline{x = 2n\pi}$$

$$x = 2n\pi + \pi - \frac{\pi}{3}$$

$$\underline{x = 2n\pi + \frac{2\pi}{3}}$$

Question 3

$$a) \int_0^{\frac{1}{4}} \frac{dx}{\sqrt{1-4x^2}}$$

$$= \int_0^{\frac{1}{4}} \frac{1}{\sqrt{1-4x^2}} dx$$

$$= \int_0^{\frac{1}{4}} \frac{1}{\sqrt{4(\frac{1}{4}-x^2)}} dx$$

$$= \int_0^{\frac{1}{4}} \frac{1}{2\sqrt{\frac{1}{4}-x^2}} dx$$

$$= \frac{1}{2} \int_0^{\frac{1}{4}} \frac{1}{\sqrt{\frac{1}{4}-x^2}} dx$$

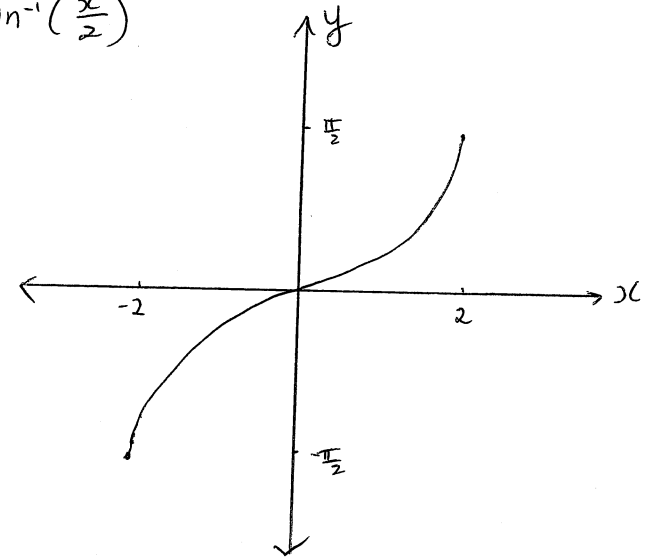
$$= \frac{1}{2} \int_0^{\frac{1}{4}} \frac{1}{\sqrt{(\frac{1}{2})^2-x^2}} dx$$

$$= \frac{1}{2} \left[\sin^{-1} \left(\frac{x}{\frac{1}{2}} \right) \right]_0^{\frac{1}{4}}$$

$$= \frac{1}{2} \left[\sin^{-1}(2x) \right]_0^{\frac{1}{4}}$$

$$= \frac{1}{2} \left[\sin^{-1}\left(\frac{1}{2}\right) - \sin^{-1}(0) \right] = \frac{1}{2} \left[\frac{\pi}{6} - 0 \right] = \frac{\pi}{12}$$

$$b) y = \sin^{-1}\left(\frac{x}{2}\right)$$



$y = \sin^{-1}x$ has domain $-1 \leq x \leq 1$

so $y = \sin^{-1}\left(\frac{x}{2}\right)$ has domain $-1 \leq \frac{x}{2} \leq 1$

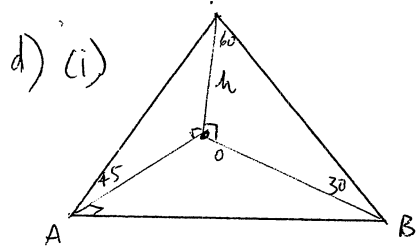
$\therefore$ Domain $-2 \leq x \leq 2$

Range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

$$c) \frac{d}{dx} [(1+x^2) \cdot \tan^{-1}x]$$

$$= \tan^{-1}x \cdot 2x + (1+x^2) \cdot \frac{1}{1+x^2}$$

$$= \underline{2x \tan^{-1}x + 1}$$



In $\triangle OAT$

$$\tan 45^\circ = \frac{h}{OA}$$

$$h = OA \tan 45$$

$$h = OA \times 1$$

$$\therefore h = OA$$

In $\triangle OBT$,

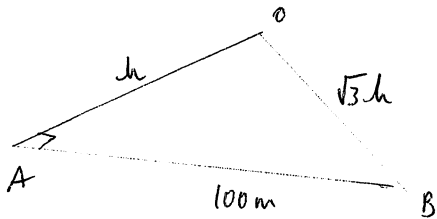
$$\tan 30 = \frac{h}{OB}$$

$$\therefore h = OB \tan 30^\circ$$

$$h = OB \times \frac{1}{\sqrt{3}}$$

$$h = \frac{OB}{\sqrt{3}}$$

$$OB = \sqrt{3}h$$



$$(\sqrt{3}h)^2 = h^2 + 100^2$$

$$3h^2 = h^2 + 100^2$$

$$2h^2 = 100^2$$

$$h^2 = \frac{100^2}{2}$$

$$h = \frac{\sqrt{100^2}}{\sqrt{2}}$$

$$\therefore h = \frac{100}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{100\sqrt{2}}{2} = \underline{\underline{50\sqrt{2}}}$$

d) (ii) $\tan \angle AOB = \frac{100}{50\sqrt{2}}$

$$= \frac{2}{\sqrt{2}}$$

$$= \frac{2\sqrt{2}}{2}$$

$$\tan \angle AOB = \sqrt{2}$$

$$\therefore \angle AOB = 54^\circ 44'$$

So B is $(180^\circ - 54^\circ 44')$ from O

so bearing is $125^\circ 16'$

Question 4

$$\begin{aligned} \text{a) } \sin^{-1}\left(\cos \frac{\pi}{3}\right) \\ &= \sin^{-1}\left(\frac{1}{2}\right) \\ &= \frac{\pi}{6} \end{aligned}$$

$$\text{b) } \int_0^{15} \frac{dx}{\sqrt{x^2+64}} \stackrel{\text{show}}{=} 2 \ln 2$$

$$\begin{aligned} \int_0^{15} \frac{1}{\sqrt{x^2+8^2}} dx &= \left[\ln \left[x + \sqrt{x^2+8^2} \right] \right]_0^{15} \\ &= \left[\ln \left[15 + \sqrt{15^2+64} \right] \right] - \left[\ln \left[0 + \sqrt{0^2+8^2} \right] \right] \\ &= \ln \left[15 + \sqrt{289} \right] - \ln \sqrt{64} \\ &= \ln \left[15 + 17 \right] - \ln 8 \\ &= \ln 32 - \ln 8 \\ &= \ln \left(\frac{32}{8} \right) \\ &= \ln 4 \\ &= \ln 2^2 \\ &= 2 \ln 2 \text{ as required.} \end{aligned}$$

$$\text{c) (i) Show that } \frac{u}{u+1} = 1 - \frac{1}{u+1}$$

$$\begin{aligned} \text{LHS } \frac{u+1-1}{u+1} \\ &= \frac{u+1}{u+1} - \frac{1}{u+1} \\ &= 1 - \frac{1}{u+1} \\ &= \text{RHS.} \end{aligned}$$

$$\text{(ii) Find } \int \frac{dx}{1+\sqrt{x}} \text{ using substitution } x=u^2 \text{ (} u \geq 0 \text{)}$$

$$\begin{aligned} x &= u^2 \\ \frac{dx}{du} &= 2u \\ dx &= 2u du \end{aligned}$$

$$= \int \frac{1}{1+\sqrt{u^2}} \cdot 2u du$$

$$= \int \frac{1}{1+u} \cdot 2u du$$

$$= \int \frac{2u}{1+u} du$$

$$= 2 \int \frac{u}{u+1} du$$

$$\text{(from above)} = 2 \int \left(1 - \frac{1}{u+1} \right) du$$

$$= 2 \left[u - \ln(u+1) \right] + C$$

$$= 2u - \ln(u+1)^2 + C$$

d) $y = \sin^{-1}(\sqrt{x})$ Show that $\frac{dy}{dx} = \frac{1}{2\sqrt{x}}$

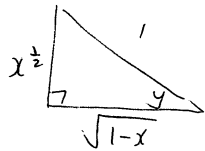
$$y = \sin^{-1}(x^{\frac{1}{2}}) \quad \text{--- ①}$$

Then $\frac{dy}{dx} = \frac{1}{\sqrt{1-x}} \times \frac{1}{2\sqrt{x}}$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x}\sqrt{1-x}}$$

Now $\frac{1}{\sin 2y} = \frac{1}{2\sin y \cos y}$

From ① $\sin y = x^{\frac{1}{2}}$
 $\cos y = \sqrt{1-x}$



$$\therefore \frac{1}{\sin 2y} = \frac{1}{2\sqrt{x}\sqrt{1-x}}$$

$$= \frac{dy}{dx}$$

OR $\sqrt{x} = \sin y$

$$x = \sin^2 y$$

$$\frac{dx}{dy} = 2\sin y \cos y$$

$$\frac{dx}{dy} = \frac{1}{2\sin y \cos y} = \frac{1}{\sin 2y}$$

Question 5

a) If $(2x+1)$ is a factor, $p(-\frac{1}{2}) = 0$

$$2(-\frac{1}{2})^3 + 7(-\frac{1}{2})^2 - (-\frac{1}{2}) - 2$$

$$= -\frac{1}{4} + \frac{7}{4} + \frac{1}{2} - 2$$

$$= 0.$$

b) (i) Show that $\sin x - \cos 2x = 2\sin^2 x + \sin x - 1$

LHS:

We know

$$\cos 2x = 1 - 2\sin^2 x$$

$$\begin{aligned} \text{so } \sin x - \cos 2x &= \sin x - (1 - 2\sin^2 x) \\ &= \sin x - 1 + 2\sin^2 x \\ &= 2\sin^2 x + \sin x - 1 \\ &= \text{R.H.S.} \end{aligned}$$

(ii) . Solve $\sin x - \cos 2x = 0$ for $0 \leq x \leq 2\pi$

Let $m = \sin x$

$$\therefore 2\sin^2 x + \sin x - 1 = 0$$

ie $2m^2 + m - 1 = 0$

$$2m^2 + 2m - m - 1 = 0$$

$$2m(m+1) - (m+1) = 0$$

$$(2m-1)(m+1) = 0$$

$$\therefore m = \frac{1}{2} \quad \text{or} \quad m = -1$$

Repeating
for m

$$\sin x = \frac{1}{2} \quad \text{or} \quad \sin x = -1$$

$$\therefore x = \frac{\pi}{6} \quad \text{or} \quad \pi - \frac{\pi}{6} \quad \text{or}$$

$$\underline{\underline{x = \frac{\pi}{6}}}, \quad \underline{\underline{\frac{5\pi}{6}}} \quad \text{or} \quad \underline{\underline{\frac{3\pi}{2}}}$$

c) If $a \cdot \cos x = 1 + \sin x$

(i) Prove that $\frac{a-1}{a+1} = t$, where $t = \tan \frac{x}{2}$

If $t = \tan \frac{x}{2}$

then $\sin x = \frac{2t}{1+t^2}$ and $\cos x = \frac{1-t^2}{1+t^2}$

so $a \cdot \cos x = 1 + \sin x$

$$a \cdot \left(\frac{1-t^2}{1+t^2} \right) = 1 + \left(\frac{2t}{1+t^2} \right)$$

$$a \left(\frac{1-t^2}{1+t^2} \right) = \frac{1+t^2}{1+t^2} + \frac{2t}{1+t^2}$$

$$a(1-t^2) = t^2 + 2t + 1$$

$$a(1-t^2) = (t+1)^2$$

$$a(1-t)(1+t) = (1+t)^2$$

$$a(1-t) = (1+t)$$

$$a - at = 1 + t$$

$$a - 1 = t + at$$

$$a - 1 = t(1+a)$$

$$\therefore t = \frac{a-1}{a+1} \text{ as required.}$$

(ii) Hence, solve $1 + \sin x = 2 \cos x$ for $0^\circ < x < 360^\circ$ to nearest degree.

$$1 + \sin x = a \cos x$$

↑

$$\text{then } a = 2$$

$$\text{and since } t = \frac{a-1}{a+1},$$

$$t = \frac{2-1}{2+1}$$

$$t = \frac{1}{3}$$

$$\text{where } t = \tan \frac{x}{2},$$

$$\frac{1}{3} = \tan \frac{x}{2}$$

$$\frac{x}{2} = \tan^{-1}\left(\frac{1}{3}\right)$$

$$\frac{x}{2} = 18^\circ 26'$$

$$\therefore x = 36^\circ 52' 11.63''$$

$$\text{so } \underline{x = 37^\circ} \text{ (to nearest degree).}$$

Question 6

$$a) \int_1^{e^3} \frac{dx}{x(9 + (\ln x)^2)}$$

$$= \int_0^3 \frac{du}{9 + u^2}$$

$$= \frac{1}{3} \left[\tan^{-1} \frac{u}{3} \right]_0^3$$

$$= \frac{1}{3} [\tan^{-1} 1 - \tan^{-1} 0]$$

$$= \frac{1}{3} \left[\frac{\pi}{4} - 0 \right]$$

$$= \underline{\underline{\frac{\pi}{12}}}$$

$$u = \ln x$$

$$\frac{du}{dx} = \frac{1}{x}$$

$$du = \frac{1}{x} dx$$

$$\therefore dx = x du$$

$$\text{when } x = e^3, \underline{u = 3}$$

$$x = 1, \underline{u = 0}$$

$$b) \quad p(x) = x^4 + bx^3 + cx^2 + dx + e$$

Since $p(x)$ is even

$$\text{Then } p(x) = x^4 + cx^2 + e$$

$$p(2) = 0 \quad \therefore 2^4 + c(2)^2 + e = 0$$

$$16 + 4c + e = 0$$

$$\underline{4c + e = -16} \quad (1)$$

When $x=3$, $p(x)$ is 55,

$$3^4 + c(3)^2 + e = 55$$

$$81 + 9c + e = 55$$

$$\underline{9c + e = -26} \quad (2)$$

(2) - (1) we have:

$$5c = -10$$

$$\therefore \underline{c = -2}$$

When $c = -2$, subst
into (1) we have

$$4(-2) + e = -16$$

$$-8 + e = -16$$

$$e = -16 + 8$$

$$e = -8$$

$$\therefore \underline{p(x) = x^4 - 2x^2 - 8}$$

$$c) \quad \tan^{-1} y = 2 \tan^{-1} x$$

$$\text{Let } \tan^{-1} x = \theta \quad \text{so } \tan \theta = x$$

$$\text{then } \tan^{-1}(y) = 2\theta$$

$$\therefore y = \tan 2\theta$$

$$y = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

$$\text{since } \tan \theta = x, \quad \underline{y = \frac{2x}{1-x^2}}$$

cii) Show that the function has no turning point.

$$\frac{dy}{dx} = \frac{(1-x^2) \cdot 2 - 2x(-2x)}{(1-x^2)^2}$$

$$= \frac{2 - 2x^2 + 4x^2}{(1-x^2)^2}$$

$$\frac{dy}{dx} = \frac{2x^2 + 2}{(1-x^2)^2}$$

for
turning pts,
 $\frac{dy}{dx} = 0$,

$$\therefore 0 = 2x^2 + 2$$

$$0 = 2(1+x^2)$$

$$x^2 + 1 = 0$$

$$x^2 = -1$$

$\therefore$ no solution and $\therefore$ no turning points.

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